

THE IDENTITY PRIMARY OBSTRUCTS THE DEGREE-SIX DESCENDANT-VIRASORO CORRESPONDENCE IN THE RESTRICTED SINE-GORDON MODEL

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ABSTRACT. The local fields of the restricted sine-Gordon model carry two descriptions. One acts on towers of form-factor polynomials by the odd local integrals of motion and by transverse generators; the other organizes descendants into Virasoro modules. These are conjectured to agree degree by degree, and at degree six the comparison sets eleven descendants of the primary ϕ_{2m} against the eleven Virasoro descendants indexed by the partitions of six, for every $m \geq 0$ and every allowed coupling. We evaluate the endpoint $m = 0$. There the primary is the identity operator and the local integrals act by commutators, so each of them annihilates it; eight of the eleven displayed descendants are products whose rightmost factor is such an integral, and therefore vanish. The remaining span has dimension at most three. The universal vacuum Verma module has degree-six dimension eleven, and its quotient by the submodule generated by the singular vector $L_{-1}v$ has dimension four. At the coupling $\xi = \sqrt{2}$, which satisfies the non-root-of-unity condition, the asserted equality consequently fails under either target. The failure is sharp and lands on the evidence advanced for the identification: that evidence is a count, matching the operators which vanish against the null vectors of the module, and at this degree eight operators vanish where seven null vectors are removed. A formulation that survives must exclude the identity sector, prescribe a limiting normalization there, or name a target of dimension at most three; the cases $m \geq 1$ are untouched by this argument.

1. INTRODUCTION

Two-dimensional conformal field theory organizes the local fields of a model into Virasoro modules, with descendants of a primary field labelled by products of negative Virasoro modes [1]. A massive integrable perturbation of such a theory retains an infinite family of local integrals of motion [2, 3], whose integrable structure in the conformal limit was developed by Bazhanov, Lukyanov and Zamolodchikov [4–6], and its local fields are instead produced by the form-factor bootstrap, as towers of polynomials in the particle rapidities [7]. The restricted sine-Gordon model, obtained by restricting the sine-Gordon solitons at a root of unity so that the ultraviolet limit is a minimal model [8–10], is the setting where the two descriptions have been compared most closely.

Smirnov proposed that the comparison is an identification, and related the null relations on the form-factor side to deformed Abelian differentials [11, 12]. Concretely, one fixes $\xi > 0$ with $q = \exp(2\pi i/\xi)$ not a root of unity, takes the primary ϕ_{2m} of weight $\Delta_{2m} = -m + m(m+1)\xi$ at central charge $c = 1 - 6/(\xi(\xi+1))$, and generates descendants with two families of operators: the odd local integrals of motion I_{2k-1} , acting on a local operator $\mathcal{O}$ by the commutator $\mathcal{I}_{2k-1}\mathcal{O} = [I_{2k-1}, \mathcal{O}]$ and carrying degree $2k-1$, and transverse generators $\mathcal{J}_{2k}$ of degree $2k$, acting on a form-factor polynomial by multiplication. The assertion is that the span of such descendants of a given degree coincides with the corresponding graded piece of the Virasoro module generated by ϕ_{2m} . Babelon, Bernard, and Smirnov developed the matching null-vector theory, identified the vacuum primary

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with the identity operator, and derived the character $(1 - p^{2m+1}) \prod_{j \geq 1} (1 - p^j)^{-1}$ of the resulting quotient [13].

The evidence Smirnov offers for the identification is a count. In the discussion following his Conjecture he observes first that the two graded spaces have the same character, and then states as the crucial check that the local operators which vanish are exactly as numerous as the null vectors of the Verma module, crediting that count to the null-vector paper [12, 13]. That check is what the present note evaluates at one point, and it is where the failure appears: at degree six in the identity sector the two numbers differ by one.

Later fermionic constructions made the correspondence explicit while keeping a sharp boundary in degree. The hidden Grassmann structure of the XXZ chain and its conformal and sine–Gordon limits supplied a fermionic basis [14–18], and the identification of that basis with the Virasoro one was carried through level six, then level eight, and then level ten, in every case working modulo the action of the local integrals of motion [17, 19, 20]. Jimbo, Miwa, and Smirnov compared fermionic and Virasoro descendants in the same modulo-integrals sense [21]. Removing that quotient is what makes the problem hard. Boos and Smirnov achieved it with nontrivial integral action through level four; at level six they report that the eleven descendants yield only nine linearly independent polynomials in their expectation-value scheme, so boundary data from ground states alone do not determine the coefficients, and they defer the case [22].

The degree-six statement left open there is the subject of this note. Written out, it asserts for every $m \geq 0$ and every allowed coupling that the span of

$$\begin{aligned} \mathcal{J}_6 \phi_{2m}, \quad \mathcal{I}_5 \mathcal{I}_1 \phi_{2m}, \quad \mathcal{J}_4 \mathcal{J}_2 \phi_{2m}, \quad \mathcal{J}_4 \mathcal{I}_1^2 \phi_{2m}, \quad \mathcal{I}_3^2 \phi_{2m}, \quad \mathcal{I}_3 \mathcal{J}_2 \mathcal{I}_1 \phi_{2m}, \\ \mathcal{I}_3 \mathcal{I}_1^3 \phi_{2m}, \quad \mathcal{J}_2^3 \phi_{2m}, \quad \mathcal{J}_2^2 \mathcal{I}_1^2 \phi_{2m}, \quad \mathcal{J}_2 \mathcal{I}_1^4 \phi_{2m}, \quad \mathcal{I}_1^6 \phi_{2m} \end{aligned} \quad (1.1)$$

equals the degree-six subspace of the Virasoro module generated by ϕ_{2m} , spanned by the eleven monomials $L_{-6} \phi_{2m}, L_{-5} L_{-1} \phi_{2m}, \dots, L_{-1}^6 \phi_{2m}$ indexed by the partitions of six.

Before a transition matrix between two eleven-element lists is sought, the quantifier deserves attention: $m \geq 0$ includes $m = 0$, and there the primary is the identity operator. Our observation is that this endpoint already settles the question, in the negative and by an exact count.

Theorem 1.1. *Take $\xi = \sqrt{2}$, so that $q = \exp(2\pi i/\xi)$ is not a root of unity, and take $m = 0$, so that $\phi_0 = 1$. Let $\mathcal{A}_0[6]$ be the span of the eleven operators in (1.1) at $m = 0$. Then*

$$\dim \mathcal{A}_0[6] \leq 3,$$

whereas the degree-six subspace of the universal Verma module $M(c, 0)$ has dimension 11 and the degree-six subspace of its quotient by the submodule generated by the singular vector $L_{-1}v$ has dimension 4. Consequently $\mathcal{A}_0[6]$ equals neither.

The mechanism is a single identity. The action of an integral of motion is a commutator, and a commutator with the identity operator vanishes, so $\mathcal{I}_{2k-1} \phi_0 = 0$ for every k . Operator products in (1.1) act on the primary from the right, so any monomial whose rightmost factor is an $\mathcal{I}$ is annihilated. Seven of the eleven end in $\mathcal{I}_1$ and an eighth, $\mathcal{I}_3^2 \phi_0$, acts first by $\mathcal{I}_3$; only the three pure- $\mathcal{J}$ monomials can survive. Three is below four, and four is already the smaller of the two conventional target dimensions, so the mismatch does not turn on which Virasoro target is meant.

This is a statement about the formulation, not a refutation of the programme it belongs to. What it shows is that the identity sector is a genuine constraint on how the correspondence may be posed: a target must be specified together with its behaviour at $m = 0$, since the operator side degenerates there for reasons that have nothing to do with the coupling or with degree six in particular. It says nothing about $m \geq 1$, where the displayed descendants need not degenerate and the eleven-against-eleven comparison retains its content.

Section 2 fixes the conventions and the two candidate targets. Section 3 proves the two facts about the identity sector. Section 4 proves Theorem 1.1. Section 5 delimits what the theorem decides and states the formulations that survive it.

2. THE DEGREE-SIX COMPARISON AND ITS TWO VIRASORO TARGETS

Fix $\xi > 0$ and set

$$q = \exp(2\pi i/\xi), \quad q \text{ not a root of unity.} \quad (2.1)$$

For $m \geq 0$, let ϕ_{2m} be the restricted sine-Gordon primary of weight

$$\Delta_{2m} = -m + m(m+1)\xi, \quad c = 1 - \frac{6}{\xi(\xi+1)}. \quad (2.2)$$

The two families of descendant-generating operators carry degrees $\deg \mathcal{I}_{2k-1} = 2k-1$ and $\deg \mathcal{J}_{2k} = 2k$. The first acts by the commutator with the local integral of motion of spin $2k-1$, a definition stated in this form by Smirnov and used unchanged since [12, 21],

$$\mathcal{I}_{2k-1}\mathcal{O} = [I_{2k-1}, \mathcal{O}], \quad (2.3)$$

and the second acts on the $2n$ -particle form-factor polynomial $L_n[\mathcal{O}](A | B)$ by

$$L_n[\mathcal{J}_{2k}\mathcal{O}](A | B) = \left(\sum_{i=1}^n A_i^{2k} - \frac{1}{2} \sum_{j=1}^{2n} B_j^{2k} \right) L_n[\mathcal{O}](A | B). \quad (2.4)$$

Write $\mathcal{A}_m[d]$ for the span, inside the space of local operators, of the degree- d descendants of ϕ_{2m} generated this way. Only (2.3) enters the argument below; (2.4) is recorded because it fixes which of the eleven monomials are pure- $\mathcal{J}$, and no property of $\mathcal{J}_{2k}$ beyond its degree is used.

Remark 2.1. The polynomial multiplier representing $\mathcal{I}_{2k-1}$ before the form-factor pairing need not vanish identically, and (2.3) does not say that it does. The two are consistent because the polynomial presentation has a null ideal: a tower all of whose form factors vanish represents the zero local operator [13]. The identity (2.3) is read in the space of local operators, which is where the asserted equality of spans lives.

The degree-six operator span $\mathcal{A}_m[6]$ is the span of the eleven monomials listed in (1.1). On the Virasoro side the comparison list is indexed by the eleven partitions of six,

$$\begin{aligned} L_{-6}\phi_{2m}, \quad L_{-5}L_{-1}\phi_{2m}, \quad L_{-4}L_{-2}\phi_{2m}, \quad L_{-4}L_{-1}^2\phi_{2m}, \quad L_{-3}^2\phi_{2m}, \quad L_{-3}L_{-2}L_{-1}\phi_{2m}, \\ L_{-3}L_{-1}^3\phi_{2m}, \quad L_{-2}^3\phi_{2m}, \quad L_{-2}^2L_{-1}\phi_{2m}, \quad L_{-2}L_{-1}^4\phi_{2m}, \quad L_{-1}^6\phi_{2m}, \end{aligned} \quad (2.5)$$

where the modes satisfy

$$[L_r, L_s] = (r-s)L_{r+s} + \frac{c}{12}(r^3-r)\delta_{r+s,0}. \quad (2.6)$$

The phrase “the Virasoro module generated by ϕ_{2m} ” admits two readings, and we keep both. Let $M(c, h)$ be the universal highest-weight module induced from a vector v with $L_0v = hv$ and $L_nv = 0$ for $n > 0$. By the Poincaré–Birkhoff–Witt theorem the ordered monomials

$$L_{-a_1}L_{-a_2}\cdots L_{-a_k}v, \quad a_1 \geq a_2 \geq \cdots \geq a_k \geq 1, \quad (2.7)$$

form a basis of $M(c, h)$, so a singular vector is a nonzero element of the universal module and becomes zero only in a quotient [23, 24]. At $h = 0$ the vector $L_{-1}v$ is singular: $L_1L_{-1}v = 2L_0v = 0$ by (2.6), and $L_nL_{-1}v = (n+1)L_{n-1}v = 0$ for $n > 1$. The second reading therefore removes it, giving the translation-invariant vacuum module

$$V^{\text{vac}}(c) = M(c, 0)/U(\text{Vir}_-)L_{-1}v, \quad (2.8)$$

which is the standard vacuum module of a chiral algebra [25].

Both readings have support in the primary literature, which is why the argument below is run against both. Smirnov states the correspondence using the words “Verma module” while separately distinguishing the reducible Verma module from the irreducible representation obtained by factoring out its singular submodule [12]. Babelon, Bernard, and Smirnov work with the physical quotient, identifying the vacuum primary with the identity operator and computing the character $(1 - p^{2m+1}) \prod_{j \geq 1} (1 - p^j)^{-1}$, whose $m = 0$ specialization $(1 - p) \prod_{j \geq 1} (1 - p^j)^{-1}$ is the character of (2.8) [13].

3. THE IDENTITY SECTOR

Lemma 3.1 (Admissible coupling). *The value $\xi = \sqrt{2}$ satisfies the condition in (2.1): with $q = \exp(2\pi i/\sqrt{2})$, there is no integer $N > 0$ with $q^N = 1$.*

Proof. Suppose $q^N = 1$ for an integer $N > 0$. Then $2\pi i N/\sqrt{2} \in 2\pi i \mathbb{Z}$, so $N/\sqrt{2} = r$ for some $r \in \mathbb{Z}$, and $r \neq 0$ because $N > 0$. Hence $\sqrt{2} = N/r \in \mathbb{Q}$, contradicting the irrationality of $\sqrt{2}$. $\square$

Lemma 3.2 (Vacuum commutator). *At $m = 0$ the primary is the identity operator, $\phi_0 = 1$, and*

$$\mathcal{I}_{2k-1}\phi_0 = 0 \quad \text{for every } k \geq 1. \quad (3.1)$$

Proof. The identification $\phi_0 = 1$ is the vacuum convention of the null-vector construction, where the primary of weight $\Delta_0 = 0$ is the identity field [13]. The conclusion is then immediate from (2.3): the identity operator commutes with every operator, so $\mathcal{I}_{2k-1}\phi_0 = [I_{2k-1}, 1] = 0$. $\square$

The weight is consistent: (2.2) gives $\Delta_0 = 0$ at $m = 0$ for every ξ , so the specialization does not depend on Lemma 3.1, and Lemma 3.1 is needed only to place the pair $(\xi, m) = (\sqrt{2}, 0)$ inside the range the assertion quantifies over.

Lemma 3.2 is not a reading imposed from outside. The same vanishing is recorded in the null-vector construction itself, by a different route: writing the vacuum primary as $\Phi_0 = 1$, Babelon, Bernard, and Smirnov list the basic null vector of its module as $\Gamma_0 = L_{-1}\Phi_0$, and identify the simplest null vector of Φ_0 with a multiple of $I_1\Phi_0$ [13]. A null vector is an operator all of whose form factors vanish, hence the zero local operator, so $\mathcal{I}_1\phi_0 = 0$ is already their statement. What (2.3) adds is that the same holds for every k , at once and without computing a form factor.

Remark 3.3. The same identity already separates the two sides at degree one, where the only operator of degree one is $\mathcal{I}_1$, so $\mathcal{A}_0[1] = 0$, while (2.7) gives $M(c, 0)[1] = \mathbb{C}L_{-1}v$ of dimension one. That mismatch is removed by passing to (2.8), whose degree-one subspace is zero. The point of the next section is that at degree six no such repair is available.

4. THE RANK OBSTRUCTION AT DEGREE SIX

Proposition 4.1 (Operator-side rank). *At $m = 0$, eight of the eleven monomials in (1.1) vanish, and*

$$\dim \mathcal{A}_0[6] \leq 3. \quad (4.1)$$

Proof. An operator product in (1.1) acts on ϕ_0 from the right, so if its rightmost factor is some $\mathcal{I}_{2k-1}$ then the whole product annihilates ϕ_0 by Lemma 3.2, whatever the remaining factors are. Table 1 records the rightmost factor of each of the eleven monomials. Seven of them end in $\mathcal{I}_1$, namely $\mathcal{I}_5\mathcal{I}_1$, $\mathcal{J}_4\mathcal{I}_1^2$, $\mathcal{I}_3\mathcal{J}_2\mathcal{I}_1$, $\mathcal{I}_3\mathcal{I}_1^3$, $\mathcal{J}_2^2\mathcal{I}_1^2$, $\mathcal{J}_2\mathcal{I}_1^4$ and $\mathcal{I}_1^6$; an eighth, $\mathcal{I}_3^2$, ends in $\mathcal{I}_3$. All eight therefore give the zero operator. The remaining three are $\mathcal{J}_6\phi_0$, $\mathcal{J}_4\mathcal{J}_2\phi_0$ and $\mathcal{J}_2^3\phi_0$, and a span of three vectors has dimension at most three. No claim about the independence of those three is needed, and none is made. $\square$

Proposition 4.2 (The two target dimensions). *At degree six,*

$$\dim M(c, 0)[6] = 11, \quad \dim V^{\text{vac}}(c)[6] = 4. \quad (4.2)$$

TABLE 1. The eleven degree-six operator monomials of (1.1) at $m = 0$, classified by the rightmost factor through which each acts on ϕ_0 . Every monomial whose rightmost factor is an integral of motion is annihilated by Lemma 3.2; the three pure- $\mathcal{J}$ monomials are the only ones not forced to vanish.

Monomial	Rightmost factor	Value at $m = 0$
$\mathcal{J}_6\phi_0$	$\mathcal{J}_6$	not forced to vanish
$\mathcal{J}_4\mathcal{J}_2\phi_0$	$\mathcal{J}_2$	not forced to vanish
$\mathcal{J}_2^3\phi_0$	$\mathcal{J}_2$	not forced to vanish
$\mathcal{I}_5\mathcal{I}_1\phi_0$	$\mathcal{I}_1$	0
$\mathcal{J}_4\mathcal{I}_1^2\phi_0$	$\mathcal{I}_1$	0
$\mathcal{I}_3\mathcal{J}_2\mathcal{I}_1\phi_0$	$\mathcal{I}_1$	0
$\mathcal{I}_3\mathcal{I}_1^3\phi_0$	$\mathcal{I}_1$	0
$\mathcal{J}_2^2\mathcal{I}_1^2\phi_0$	$\mathcal{I}_1$	0
$\mathcal{J}_2\mathcal{I}_1^4\phi_0$	$\mathcal{I}_1$	0
$\mathcal{I}_1^6\phi_0$	$\mathcal{I}_1$	0
$\mathcal{I}_3^2\phi_0$	$\mathcal{I}_3$	0

Proof. By (2.7) the monomials $L_{-a_1} \cdots L_{-a_k} v$ with $a_1 \geq \cdots \geq a_k \geq 1$ and $\sum_i a_i = 6$ form a basis of $M(c, 0)[6]$, so its dimension is the number of partitions of six, $p(6) = 11$.

For the quotient, the degree-six subspace of $U(\text{Vir}_-)L_{-1}v$ is spanned by the vectors $XL_{-1}v$ with $X \in U(\text{Vir}_-)$ of degree five, and $U(\text{Vir}_-)$ in degree five is spanned by ordered monomials $L_{-a_1} \cdots L_{-a_k}$ with $a_1 \geq \cdots \geq a_k \geq 1$ and $\sum_i a_i = 5$. For such a monomial the product $L_{-a_1} \cdots L_{-a_k} L_{-1}v$ is again ordered, because $a_k \geq 1$; it is therefore already one of the basis vectors (2.7), namely the one indexed by the partition of six obtained by appending a part equal to 1. That map from partitions of five to partitions of six is a bijection onto those partitions of six having at least one part equal to 1, so the spanning set consists of $p(5) = 7$ distinct basis vectors of $M(c, 0)[6]$ and is in particular linearly independent. Hence $\dim(U(\text{Vir}_-)L_{-1}v)[6] = 7$ exactly, and

$$\dim V^{\text{vac}}(c)[6] = p(6) - p(5) = 11 - 7 = 4.$$

The surviving quotient basis is indexed by the four partitions of six with no part equal to one: $6, 4+2, 3+3$, and $2+2+2$. The same number is the coefficient of p^6 in the character $(1-p) \prod_{j \geq 1} (1-p^j)^{-1}$ of [13] at $m = 0$. $\square$

Proof of Theorem 1.1. By Lemma 3.1 the coupling $\xi = \sqrt{2}$ satisfies (2.1), so the pair $(\xi, m) = (\sqrt{2}, 0)$ is one the assertion quantifies over. Proposition 4.1 bounds $\dim \mathcal{A}_0[6]$ by 3, and Proposition 4.2 gives the two target dimensions 11 and 4. Since $3 < 4 < 11$, the space $\mathcal{A}_0[6]$ has dimension different from both, so it equals neither. $\square$

Corollary 4.3 (The counting check fails by one). *At $m = 0$ and degree six, the number of displayed monomials forced to vanish is eight, while the number of null vectors removed from the Verma module at that degree is $p(5) = 7$. The two counts that Smirnov's crucial check requires to agree therefore differ by one.*

Proof. Proposition 4.1 exhibits eight monomials of (1.1) annihilated by Lemma 3.2. The proof of Proposition 4.2 shows that the degree-six part of $U(\text{Vir}_-)L_{-1}v$, the submodule of null vectors at $m = 0$, has dimension exactly $p(5) = 7$. If the span of the eleven monomials were to fill the quotient (2.8), whose degree-six dimension is $11 - 7 = 4$, then at most seven of them could vanish. Eight do. $\square$

Stated this way the obstruction is not a mismatch with a convention but a failure of the evidence advanced for the correspondence. Smirnov supports the conjecture by observing that the operators which vanish are as numerous as the null vectors [12]. At $m = 0$ they are more numerous, and the surplus is exactly the one dimension by which the operator span falls short of the vacuum quotient. The discrepancy is small, which is part of why it survives a character comparison: characters agree degree by degree only when the vanishing is counted correctly, and at this point it is not.

The theorem is stronger than exhibiting a linear dependence among the eleven operator candidates. A dependence would reduce the operator side to some dimension below eleven and leave open whether the physical quotient absorbs the difference. Here the shortfall survives the quotient: even after the vacuum singular vector and all of its descendants are removed, the target retains four dimensions against an operator span of at most three.

Remark 4.4 (On computation). No step above is computer assisted. The classification in Table 1 is an inspection of eleven written monomials, and the two dimension counts are the partition numbers $p(6) = 11$ and $p(5) = 7$; both can be checked by hand in a few minutes, and both are stated in the text rather than delegated. We did check the transcription of the eleven monomials, the eight forced zeros, and the two partition counts with an independent short program, but it certifies a transcription rather than an argument, and the proof does not rest on it.

5. SCOPE, AND THE FORMULATIONS THAT SURVIVE

Theorem 1.1 rests on three readings of the assertion, and it is worth being explicit that they are readings. The quantifier $m \geq 0$ is taken to include $m = 0$; the primary there is the identity operator without a limiting renormalization; and the operator $\mathcal{I}_{2k-1}$ acts by the commutator (2.3). Each is what the statement says, and the three are mutually consistent, which is what makes the finite argument exact. Together they are also the whole content of the hypothesis: the argument uses no property of $\mathcal{J}_{2k}$, no value of the central charge, and no feature of degree six beyond the list (1.1) itself.

What the theorem does not establish is that the authors of the correspondence intended this endpoint. The primary sources mix the two levels of description deliberately. Smirnov phrases the proposal with “Verma module” while elsewhere separating that module from its irreducible quotient, and discusses vanishing local operators in the same passage [12]; Babelon, Bernard, and Smirnov work throughout with the physical quotient and its character [13]. Their printed coupling, weight, deformation, and integral normalizations do not admit a verified term-by-term transport into every later convention, so we have not attempted one. The argument above needs none, since it uses only the commutator identity and the displayed list.

Nor does the theorem constrain $m \geq 1$. There ϕ_{2m} is not the identity, $\mathcal{I}_{2k-1}\phi_{2m}$ need not vanish, and the eleven-against-eleven comparison keeps its content; the rank shortfall proved here is a property of the identity sector alone.

Several mathematically distinct formulations survive, and they are genuinely different problems rather than restatements of one another:

- (1) restrict the assertion to $m \geq 1$, and establish the degree-six transition matrix for generic non-root-of-unity coupling;
- (2) prescribe a renormalized $m \rightarrow 0$ limit in which the displayed descendants are rescaled before the primary is specialized, so that the identity sector is approached rather than evaluated;
- (3) define the null ideal of the form-factor presentation intrinsically, and prove that the resulting quotient agrees with the appropriate Virasoro singular quotient; or
- (4) replace equality by an equality modulo the local integrals of motion, keeping explicit track of the discarded directions.

The fourth is the sense in which the correspondence is already established through level six and beyond [17, 19–21], and it is consistent with Theorem 1.1 for a concrete reason: quotienting by

the integrals of motion removes precisely the directions whose absence the theorem counts. Exact expectation values without that quotient are known only at low level: at level two for the Virasoro descendant $L_{-2}\bar{L}_{-2}\Phi$ [26], and at level four once the integrals of motion act nontrivially [22]; see also the study of one-point functions of descendants in the sine-Gordon model [27]. Degree six is the next case, and it is where the present question begins.

One repair is not available. An unspecified “physical quotient” cannot rescue the literal statement by stipulation, because the theorem rules out both quotients that the sources actually define. Any smaller target must be specified independently, and its degree-six dimension and the operator map into it computed; a target of dimension at most three is not excluded by this argument, but neither is it supplied by it.

6. CONCLUSION

The identity sector is a sharp test of any proposed correspondence between integral-of-motion descendants and Virasoro descendants, because the operator side degenerates there for a reason internal to the definition: the action is a commutator, and the identity operator commutes with everything. Applied to the degree-six list, the test is decisive. Eight of the eleven descendants vanish, the span drops to dimension at most three, and the two conventional Virasoro targets retain dimensions eleven and four. At $\xi = \sqrt{2}$ and $m = 0$, an allowed pair, the asserted equality fails.

What makes the failure informative rather than merely negative is where it lands. The correspondence is supported by a count of vanishing operators against null vectors, and [Corollary 4.3](#) shows that count to be wrong by one at this degree and this primary. A single dimension is small enough to survive a character comparison and large enough to defeat an equality of spans, which is the gap between the two kinds of evidence.

The consequence is a constraint on how the correspondence is posed rather than on whether it holds. A degree-six transition matrix can be claimed only once the target module and the treatment of the endpoint $m = 0$ are stated together, since the two interact: the quotient that repairs degree one, as in [Remark 3.3](#), does not repair degree six. The natural next step is the first of the surviving formulations above — the assertion restricted to $m \geq 1$, where the eleven-against-eleven comparison is not degenerate and where the reported linear dependence among the level-six expectation-value polynomials [22] suggests the answer is not simply affirmative either. Settling the intended convention at $m = 0$ would determine whether this note records a counterexample or a needed hypothesis.

DATA AND CODE AVAILABILITY

The proof is analytic and finite, and no data underlie it. The short program described in [Remark 4.4](#), which independently re-checks the transcription of the eleven monomials, the eight forced zeros, and the partition counts $p(6) = 11$ and $p(5) = 7$, is available from the author on request. It has no bearing on the validity of the argument, which is given in full in [Section 4](#).

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